Power

File: power.kicad_sch

MCU

File: mcu.kicad_sch

Ethernet client port

File: eth_client.kicad_sch

Ethernet I/O port

File: eth_io.kicad_sch

Memories and display

File: periph.kicad_sch

Field digital I/O

File: io_dig.kicad_sch

Field power, pump, relays, USB

File: io_pwr.kicad_sch

Analog inputs and temperatures

File: io_ana.kicad_sch

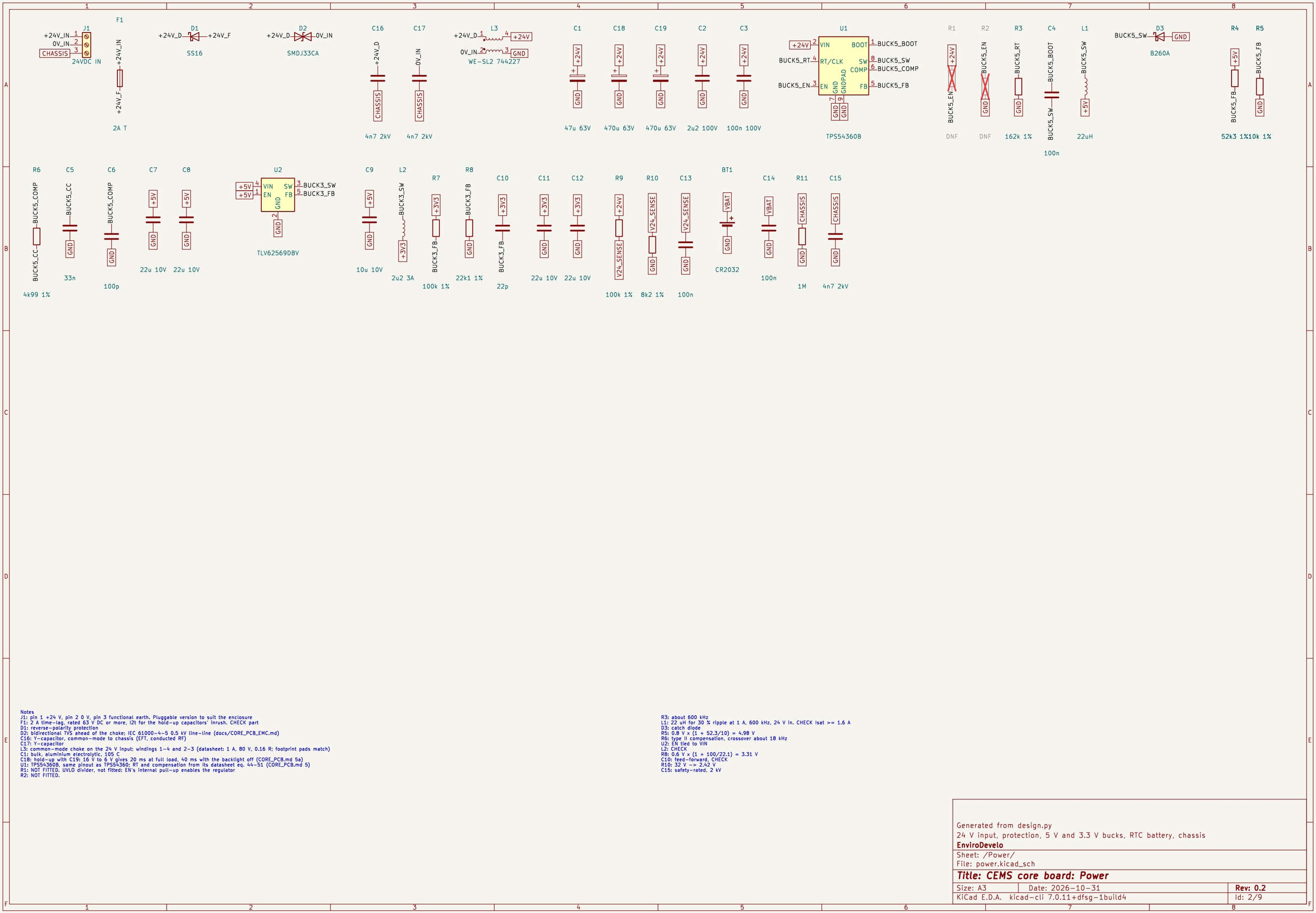
CEMS core board, rev 0.1 schematic (draft for review)
Two separate Ethernet networks: client port = STM32H573 MAC + LAN8742A (J2); I/O port = W5500 (J3). I/O boards only.
24 VDC in; DIN-rail enclosure behind the RVT70 display (FFC J4).
Pin map and design notes: docs/CORE_PCB.md. Parts marked CHECK in the sheet notes need datasheet confirmation before layout.

Design spec: docs/CORE_PCB.md
Generated from hardware/core_pcb/tools/design.py; do not edit by hand
EnviroDevelo

Sheet: /
File: cems_core.kicad_sch

Title: CEMS core board

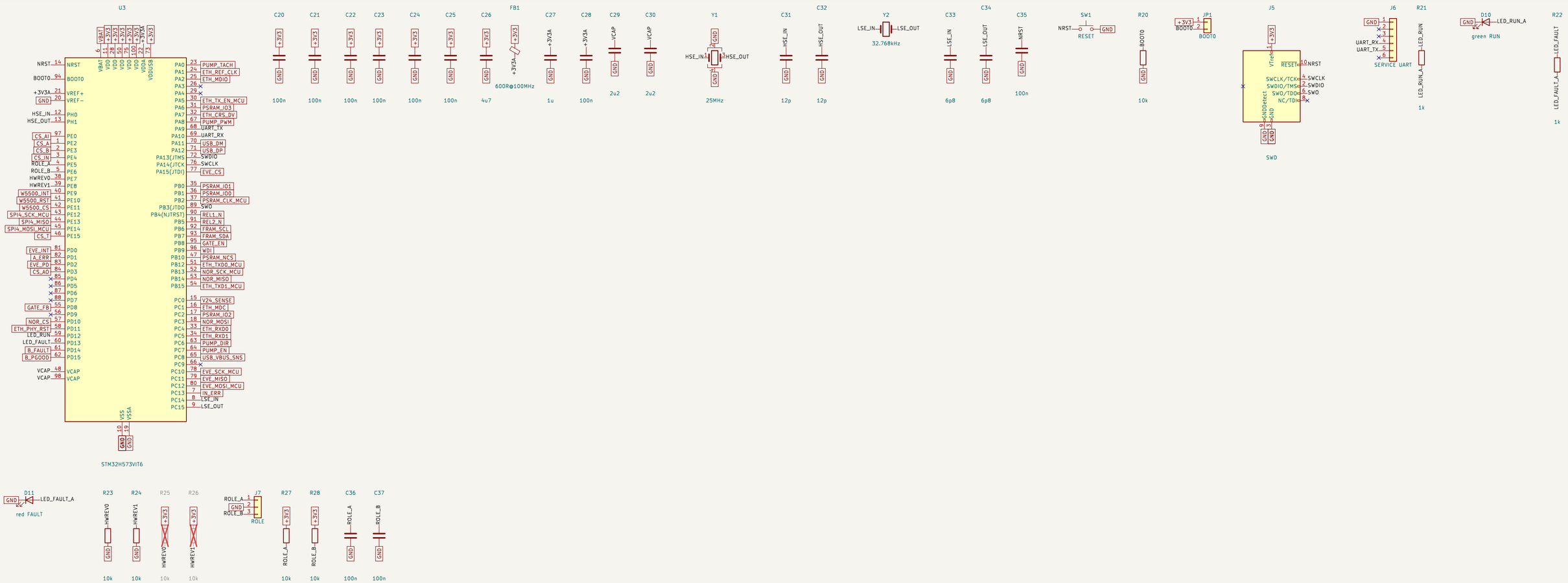
Size: A4	Date: 2026-10-31	**Rev: 0.2**
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Notes
J1: pin 1 +24 V, pin 2 0 V, pin 3 functional earth. Pluggable version to suit the enclosure
F1: 2 A time-lag, rated 63 V DC or more, I2t for the hold-up capacitors' inrush. CHECK part
D1: reverse-polarity protection
D2: bidirectional TVS ahead of the choke; IEC 61000-4-5 0.5 kV line-line (docs/CORE_PCB_EMC.md)
C16: Y-capacitor, common-mode to chassis (EFT, conducted RF)
C17: Y-capacitor
L3: common-mode choke on the 24 V input; windings 1-4 and 2-3 (datasheet: 1 A, 80 V, 0.16 R; footprint pads match)
C1: bulk, aluminium electrolytic, 105 C
C18: hold-up with C19: 16 V to 6 V gives 20 ms at full load, 40 ms with the backlight off (CORE_PCB.md 5a)
U1: TPS54360B, same pinout as TPS54360; RT and compensation from its datasheet eq. 44-51 (CORE_PCB.md 5)
R1: NOT FITTED. UVLO divider, not fitted: EN's internal pull-up enables the regulator
R2: NOT FITTED.
R3: about 600 kHz
L1: 22 uH for 30 % ripple at 1 A, 600 kHz, 24 V in. CHECK Isat >= 1.6 A
D3: catch diode
R5: 0.8 V x (1 + 52.3/10) = 4.98 V
R6: type II compensation, crossover about 18 kHz
U2: EN tied to VIN
L2: CHECK
R8: 0.6 V x (1 + 100/22.1) = 3.31 V
C10: feed-forward, CHECK
R10: 32 V -> 2.42 V
C15: safety-rated, 2 kV

Generated from design.py
24 V input, protection, 5 V and 3.3 V bucks, RTC battery, chassis
EnviroDevelo

Sheet: /Power/		
File: power.kicad_sch		
Title: CEMS core board: Power		
Size: A3	Date: 2026-10-31	**Rev: 0.2**
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Notes
U3: STM32H563VIT6 is pin-compatible
C20: one per VDD pin, VDDUSB included
C21: one per VDD pin, VDDUSB included
C22: one per VDD pin, VDDUSB included
C23: one per VDD pin, VDDUSB included
C24: one per VDD pin, VDDUSB included
C25: one per VDD pin, VDDUSB included

FB1: VDDA / VREF+ filter
C29: VCAP, CHECK value and count against the H5 datasheet
Y1: HSE, CL 10 pF class, CHECK against the DK
C31: CHECK for the crystal's CL
Y2: CL 6 pF, ESR per AN2867
C33: AN2867, CHECK for the crystal
JP1: fit a jumper for the ST bootloader

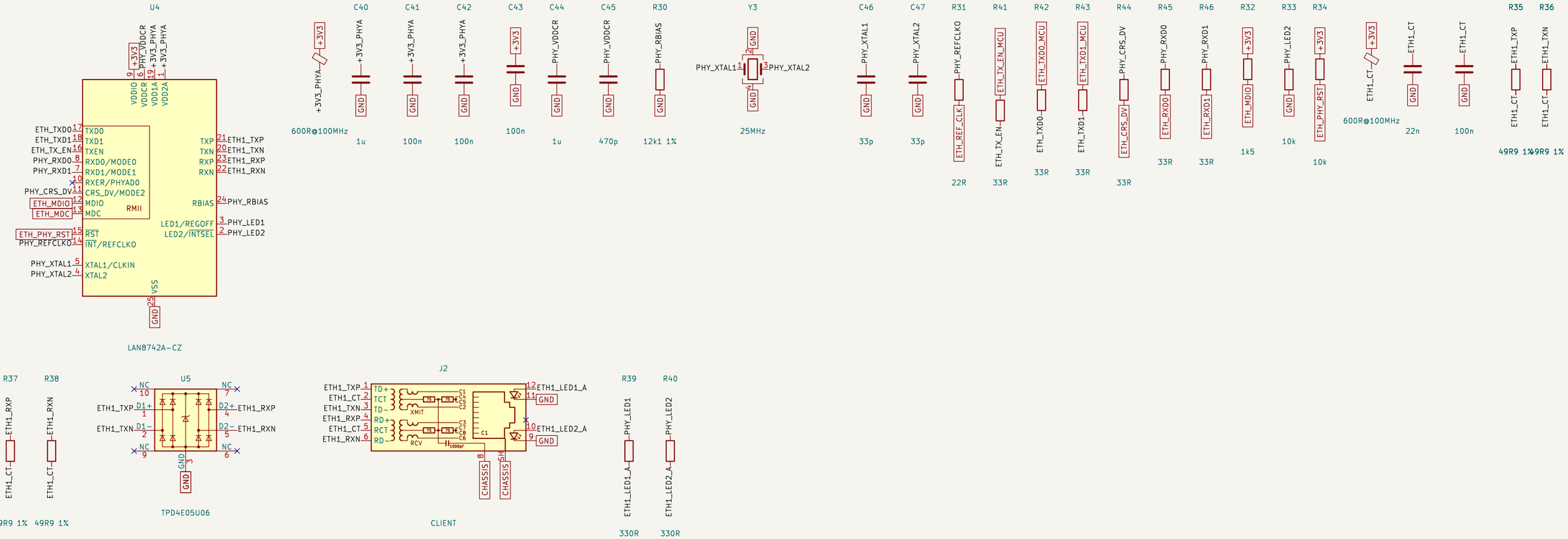
J6: FTDI TTL-232R-3V3 order: 1 GND, 2 CTS, 3 VCC (not connected), 4 cable TXD, 5 cable RXD, 6 RTS
R23: hardware revision 0: both pulled down
R25: NOT FITTED.
R26: NOT FITTED.
J7: shunt with pull tab; silkscreen 1–2 CORE / 2–3 I/O NODE / none = OFF: reachable only with the housing open
C36: ESD, at J7

Generated from design.py
STM32H573VIT6, clocks, reset, boot, SWD, service UART, LEDs
EnviroDevelo

Sheet: /MCU/
File: mcu.kicad_sch

Title: CEMS core board: MCU

Size: A2	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/9



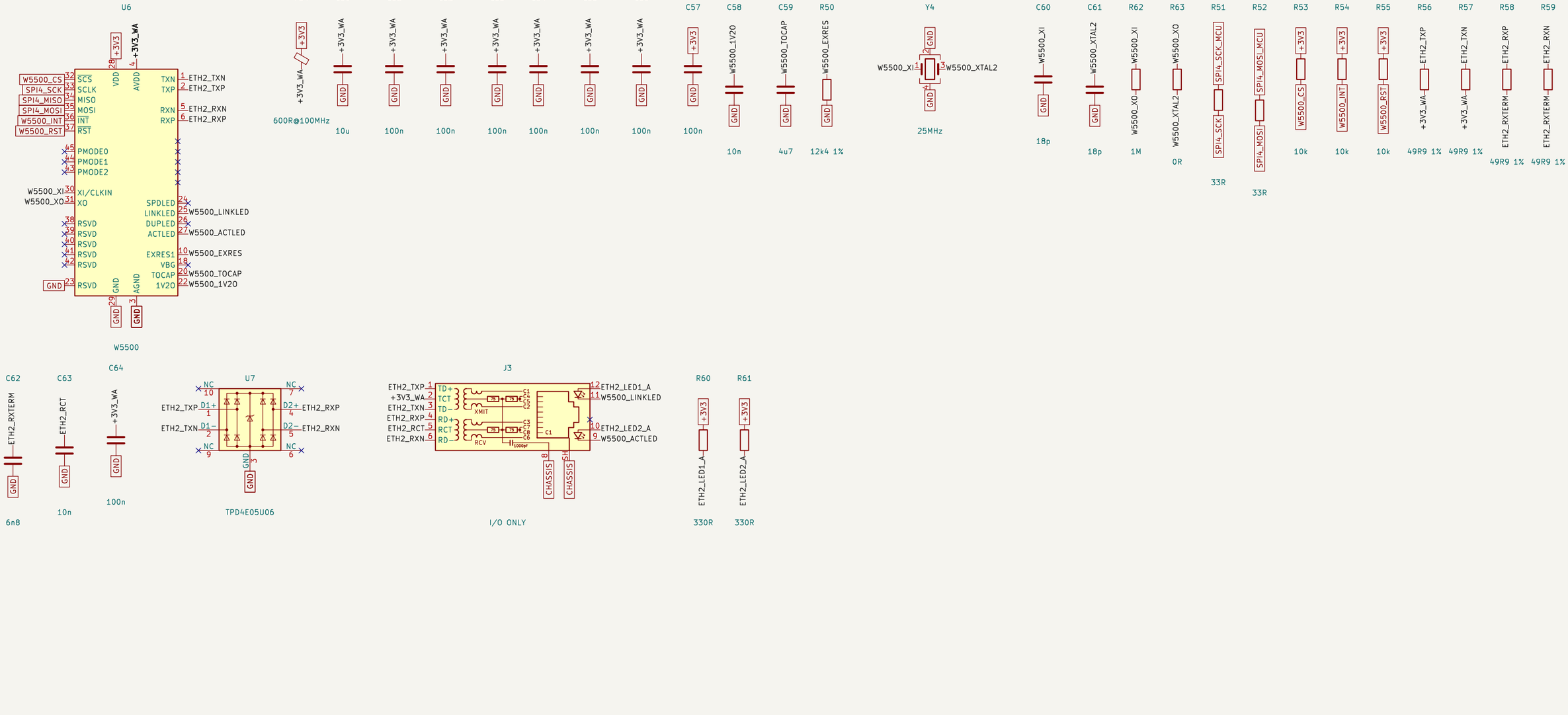
Notes
U4: straps: MODE[2:0] internal pull-ups = all capable; PHYAD0 internal pull-down = address 0; nINTSEL pulled low (R33) = pin 14 is REFCLKO, 50 MHz to the MCU; REGOFF internal pull-down = regulator on
C41: at pin 1
C42: at pin 19
C43: VDDIO
C44: VDDCR, with C45
Y3: CL 20 pF, +/-50 ppm total (LAN8742A datasheet)
C46: for CL 20 pF with about 4 pF stray
R31: series, at the PHY
R41: series, at the MCU
R42: series, at the MCU
R43: series, at the MCU
R44: series, at the PHY
R45: series, at the PHY
R46: series, at the PHY
R33: strap nINTSEL = 0 (REFCLKO mode)
R34: PHY runs if the firmware leaves PD11 as input
U5: flow-through: route each line across pins 1/10, 2/9, 4/7, 5/6
J2: client network
R39: LED1 = link/activity, active high (REGOFF strap low)
R40: LED2 = speed, active high (nINTSEL strap low)

Generated from design.py
LAN8742A on RMII, client RJ45 (J2)
EnviroDevelo

Sheet: /Ethernet client port/
File: eth_client.kicad_sch

Title: CEMS core board: Ethernet client port

Size: A3	Date: 2026-10-31	**Rev: 0.2**
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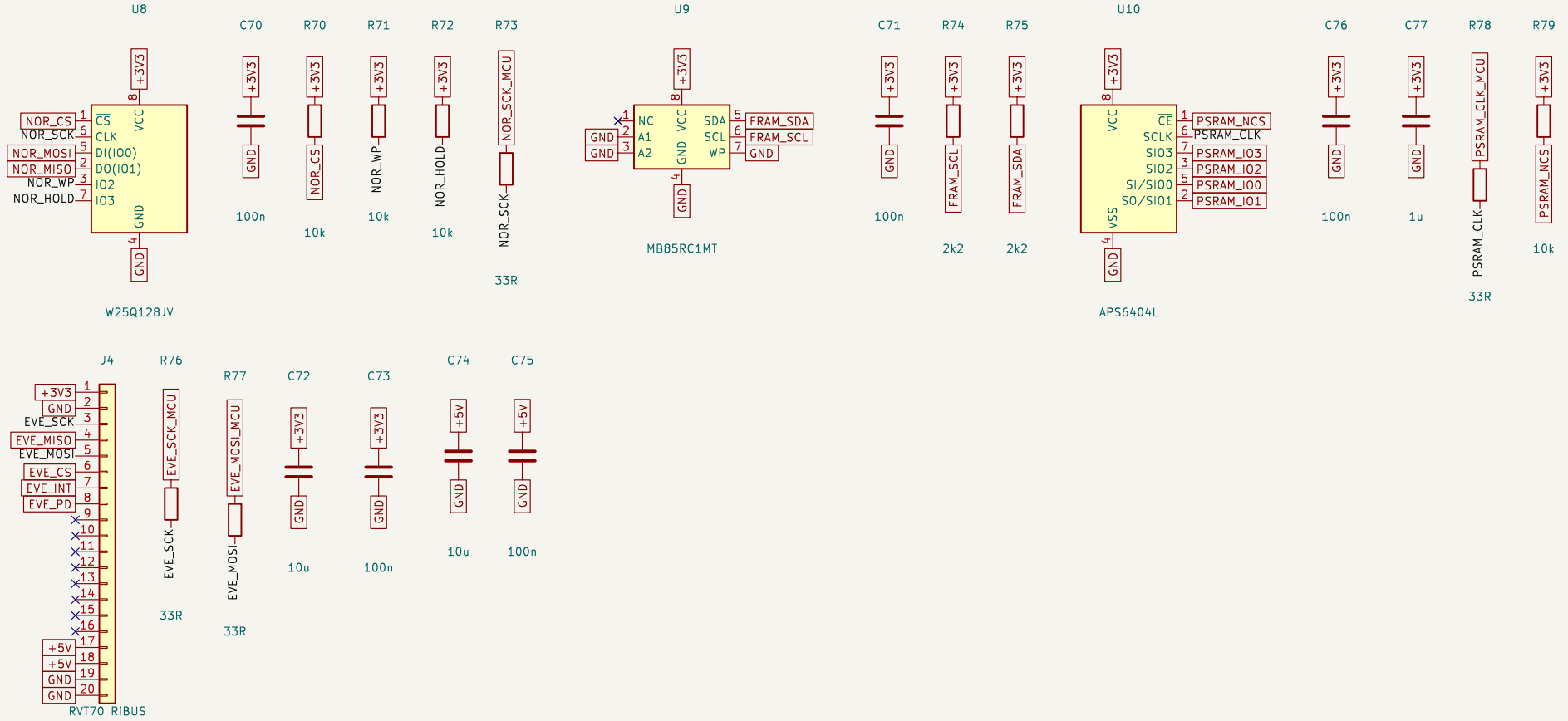


Notes
U6: PMODE[2:0] internal pull-ups = all capable, auto-negotiation; RSVD pin 23 to ground, RSVD 38–42 NC; VBG open. No auto-MDIX: the I/O board's LAN8742A does HP auto-MDIX, so a straight cable works
C51: one per AVDD pin
C52: one per AVDD pin
C53: one per AVDD pin
C54: one per AVDD pin
C55: one per AVDD pin
C56: one per AVDD pin
C57: VDD pin 28
Y4: own crystal, CL 18 pF, +/-30 ppm (W5500 datasheet); separate from the MCU's HSE

C60: W5500 reference circuit
R62: W5500 reference circuit
R63: drive-level option, W5500 reference circuit
R51: series, at the MCU
R52: series, at the MCU
C64: TX centre tap
J3: I/O boards only; silkscreen I/O ONLY
R60: link LED; W5500 LED outputs sink (active low)
R61: activity LED

Generated from design.py
W5500 on SPI4, I/O RJ45 (J3), I/O boards only
EnviroDevelo

Sheet: /Ethernet I/O port/		
File: eth_io.kicad_sch		
Title: CEMS core board: Ethernet I/O port		
Size: A3	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/9



Notes
U8: event log and data logger (H3)
R73: series, at the MCU
U9: 128 KB FRAM (H2), I2C address 0x50. A1 = A2 = 0, WP low = writes enabled (pinout per datasheet)
U10: 8 MB QSPI PSRAM on OCTOSPI1, up to about 100 MHz: keep it next to the MCU, traces under 30 mm and roughly matched
C76: at U10 VDD
R78: series, at the MCU
R79: keeps the PSRAM deselected through reset
J4: RiBUS pinout from the RVT70HSBNWC00 V1.1A datasheet; 9–12 (GPIO, audio, QSPI) left open; 13–16 NC. CHECK contact side against the cable at

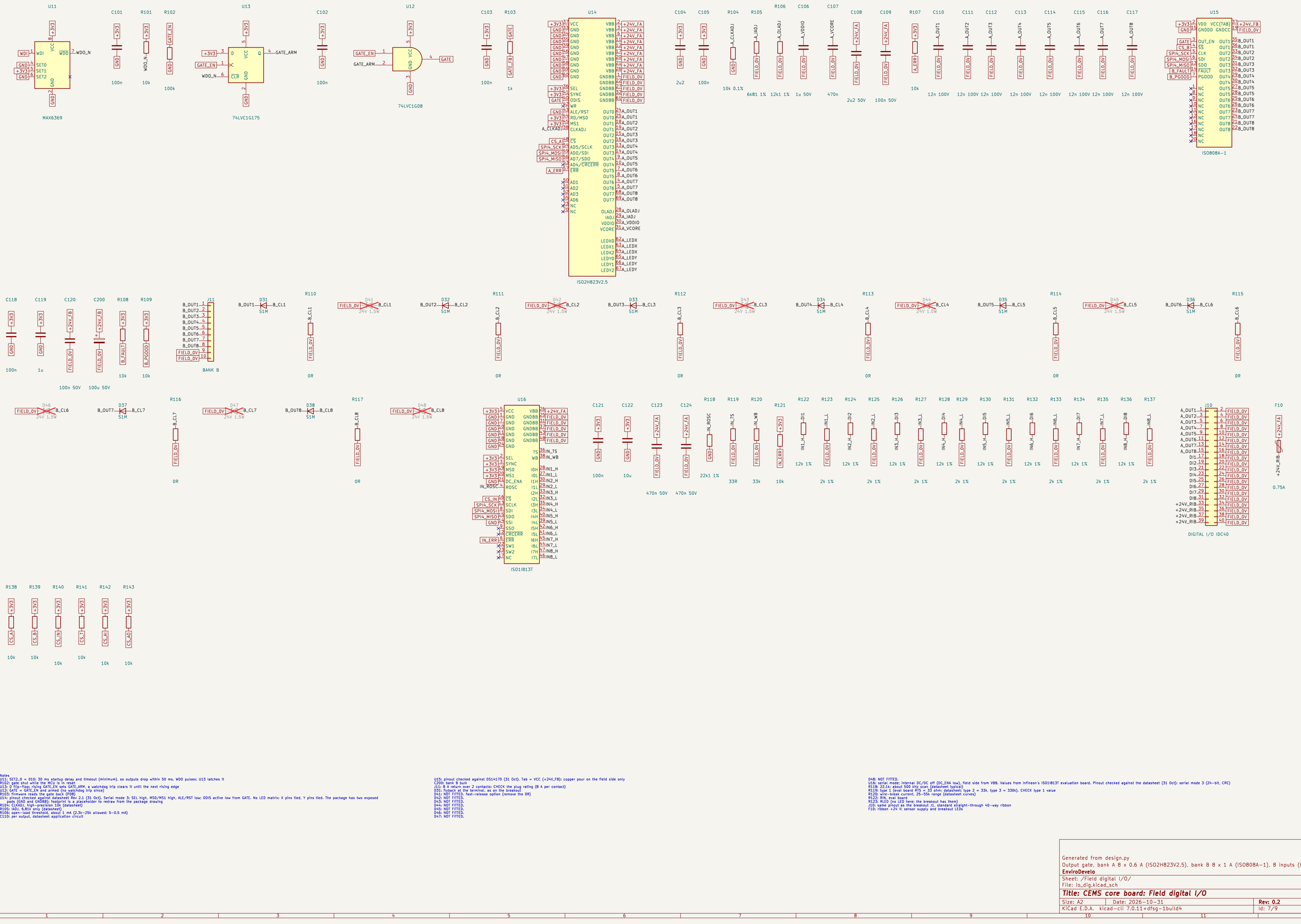
mechanical layout
R76: series, at the MCU
R77: series, at the MCU
C72: at J4
C73: at J4
C74: at J4, backlight
C75: at J4

Generated from design.py
NOR flash, FRAM, RVT70 display FFC (J4)
EnviroDevelo

Sheet: /Memories and display/
File: periph.kicad_sch

Title: CEMS core board: Memories and display

Size: A4	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/9



Notes
U11: SET2..0 = 010: 30 ms startup delay and timeout (minimum), so outputs drop within 50 ms. WDO pulses: U13 latches it
R102: gate shut while the MCU is in reset
U13: D flip-flop; rising GATE_EN sets GATE_ARM, a watchdog trip clears it until the next rising edge
U12: GATE = GATE_EN and armed (no watchdog trip since)
R103: firmware reads the gate back (PD8)
U14: pinout checked against datasheet Rev 2.1 (31 Oct). Serial mode 3: SEL high, MSO/MS1 high, ALE/RST low; ODIS active low from GATE. No LED matrix: X pins tied, Y pins tied. The package has two exposed pads (GND and GND8B): footprint is a placeholder to redraw from the package drawing
R104: CLKADJ, high-precision 10k (datasheet)
R105: IADJ, 6.81k only (datasheet)
R106: open-load threshold, about 1 mA (2.3k–25k allowed: 5–0.5 mA)
C110: per output, datasheet application circuit

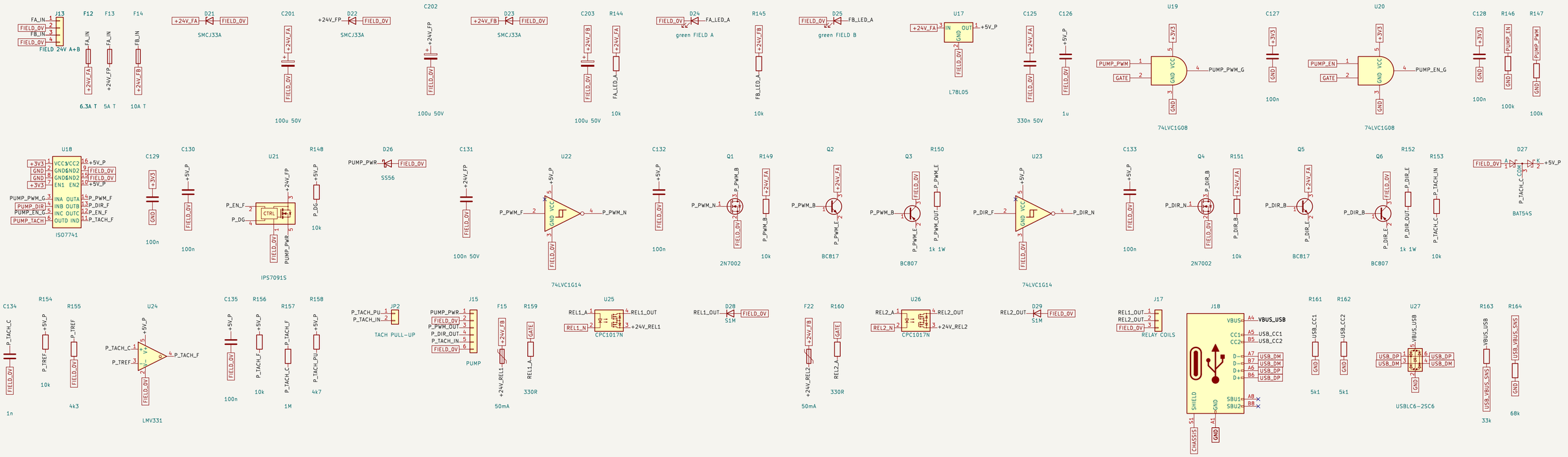
U15: pinout checked against DS14170 (31 Oct). Tab = VCC (+24V_FB): copper pour on the field side only
C200: bank B bulk
J11: B A return over 2 contacts: CHECK the plug rating (8 A per contact)
D31: flyback at the terminal, as on the breakout
D41: NOT FITTED, fast-release option (remove the 0R)
D42: NOT FITTED.
D43: NOT FITTED.
D44: NOT FITTED.
D45: NOT FITTED.
D46: NOT FITTED.
D47: NOT FITTED.

D48: NOT FITTED.
U16: serial mode: internal DC/DC off (DC_ENA low), field side from VBB. Values from Infineon's ISO1B13T evaluation board. Pinout checked against the datasheet (31 Oct): serial mode 3 (24-bit, CRC)
R118: 22.1k: about 500 kHz scan (datasheet typical)
R119: type 1 (eval board RTS = 33 ohm; datasheet: type 2 = 33k, type 3 = 330k). CHECK type 1 value
R120: wire-break current, 25–55k range (datasheet curves)
R122: RIN, eval board
R123: RLED (no LED here: the breakout has them)
J10: same pinout as the breakout J1, standard straight-through 40-way ribbon
F10: ribbon +24 V: sensor supply and breakout LEDs

Generated from design.py
Output gate, bank A 8 x 0.6 A (ISO2H823V2.5), bank B 8 x 1 A (ISO808A-1), 8 inputs (ISO1B13T)
EnviroDevelo
Sheet: /Field digital I/O/
File: io_dig.kicad_sch

Title: CEMS core board: Field digital I/O

Size: A2	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/9



Notes
J13: one 4-way plug for both field supplies (12 A per contact): A = bank A. Inputs, pump (about 10 A); B = bank B and relays (about 8 A). Link A and B in the plug for a single supply
F12: bank A. Inputs, ribbons
F13: pump
F14: bank B. relays
D21: unidirectional TVS; surge clamp, and blows the fuse on reverse polarity
U17: pump logic 5 V, field side
U18: pinout checked (31 Oct)

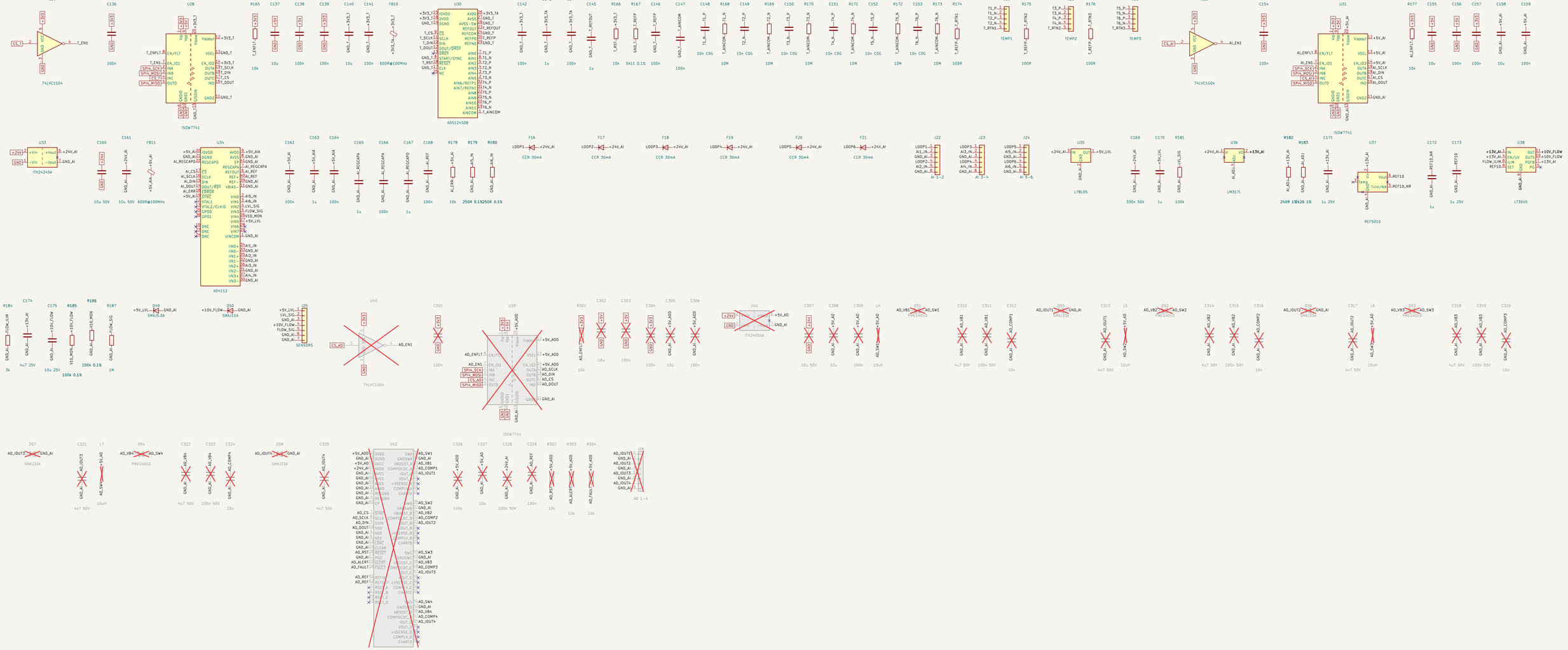
U21: about 4 A pump supply with current limit; can PWM a brushed motor to 1 kHz. CHECK rating once the 107Z current is known
R148: diagnostic, test point only
D26: motor freewheel
R150: pump PWM: 24 V push-pull, 20 Hz–25 kHz, 1k makes the output survive a short to 0 V or 24 V for ever (24 mA, 0.6 W). CHECK the 107Z input current (high level = 24 V − 1k x I)
R152: pump DIR: 24 V push-pull, 20 Hz–25 kHz, 1k makes the output survive a short to 0 V or 24 V for ever (24 mA, 0.6 W). CHECK the 107Z input current (high level = 24 V − 1k x I)
D27: clamp to the 5 V rail
R155: 1.5 V threshold

U24: CHECK pin order for the SOT-23-5 variant
R157: hysteresis
JP2: fit for an open-collector tacho
J15: one plug for the whole pump: power, 0 V, PWM, DIR, tacho, signal 0 V (Thomas 107Z: signal ground must be field 0 V)
F15: per-channel limit: a shorted coil wire trips only its own PTC
U25: 60 V, 400 mA PhotoMOS: rides out a short until its PTC trips. CHECK with a 10 s short test
R164: 5 V -> 3.4 V

Generated from design.py
Field supplies A and B, pump switch / PWM / DIR / tacho, 2 relay coil outputs, USB-C
EnviroDevelo
Sheet: /Field power, pump, relays, USB/
File: io_pwr.kicad_sch

Title: CEMS core board: Field power, pump, relays, USB

Size: A2	Date: 2026-10-31	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/9



Notes

U28: VSEL to GND2 → 3.3 V out (1/4 mA)
U30: ENALL_DAC1 high (VIO level) enables the DC/DC; reads low on a fault
U30: pinout checked (SM45600, 31 Oct); internal 4.096 MHz clock (CLK to DGND), START low (conversions by command), DOUT/DRDY on one pin
C146: REFOUT to REFCOM
C147: CHECK AINCOM and the TC bias against the Ia-node schematic
C148: differential filter
R168: RTD return
R174: RTD return to the reference resistor
J19: T1 +/– T2 +/– RTD return
J20: T3 +/– T4 +/– RTD return
J21: T5 +/– T6 +/– RTD return
U31: VSEL to VSDOUT = 5 V out (110 mA)
R177: EN/FLT high (VIO level) enables the DC/DC; reads low on a fault
U33: loop power-supply input, 9–36 V nominal; check Vout and Vin pin-outs
U34: internal oscillator and 2.5 V ref; VINA 10 V monitor; VIN5 5 V monitor; VIN2 level; VIN3 mass flow; VIN4 10 V monitor; VINS 5 V monitor, Pinout checked (31 Oct): CHECK the exposed pad size against the CP-40 drawing
R178: 4–20 mA shunt → 1–5 V; takes the 30 mA limit for ever (0.23 W)
R180: 4–20 mA shunt → 1–5 V; takes the 30 mA limit for ever (0.23 W)
J22: per channel: 24 V loop, IA, 0 V (2-wire or 4-wire transmitters); CHECK the 30 mA option, its 2.5 V headroom and the AD4112 IIN abs max
J22: per channel: 24 V loop, IA, 0 V (2-wire or 4-wire transmitters)
J23: per channel: 24 V loop, IA, 0 V (2-wire or 4-wire transmitters)
J24: per channel: 24 V loop, IA, 0 V (2-wire or 4-wire transmitters)
U35: level sensor 5 V, internal current limit
R181: defined level when unplugged
U36: 240R sets 10 mA minimum; 26 copper on the VO pins
R183: 1.25 V (1 + 226Ω/240) = 2.5 V
U38: SET driven by the REF5010 (no RSET); CHECK PGFB tie when PG is unused
R184: about 50 mA current limit

R185: 10 V monitor for ratiometric correction
D49: clamps +5V_LVL
D50: clamps +10V_FLOW (12.2 V min breakdown)
J25: level (DFRobot SEN0205): 5 V, SIG, 0 V: mass flow (HAMS10VHV): +10 V, OUT, 0 V
C301: NOT FITTED.
C302, NOT FITTED.
U39: NOT FITTED, VSEL to VSDOUT = 5 V out (DVDD)
R301: NOT FITTED, EN/FLT high (VIO level) enables the DC/DC; reads low on a fault
C303: NOT FITTED.
C304: NOT FITTED.
C305: NOT FITTED.
C306: NOT FITTED.
U41: NOT FITTED, AVCC for the four boost converters
C307: NOT FITTED.
C308: NOT FITTED.
C309: NOT FITTED.
L4: NOT FITTED, boost inductor, as the datasheet test circuit
C310: NOT FITTED.
C311: NOT FITTED.
C312: NOT FITTED, COMPDCDC
D55: NOT FITTED, output clamp; CHECK what IOUT survives when 24 V is wired onto it
C313: NOT FITTED.
L5: NOT FITTED.
D52: NOT FITTED.
C314: NOT FITTED.
C315: NOT FITTED.

C316: NOT FITTED.
D56: NOT FITTED.
C317: NOT FITTED.
L6: NOT FITTED.
D53: NOT FITTED.
C318: NOT FITTED.
C319: NOT FITTED.
C320: NOT FITTED.
D57: NOT FITTED.
C321: NOT FITTED.
L7: NOT FITTED.
D54: NOT FITTED.
C322: NOT FITTED.
C323: NOT FITTED.
C324: NOT FITTED.
D58: NOT FITTED.
C325: NOT FITTED.
U42: NOT FITTED, 4 × 4–20 mA, short and open-loop detect. Pin numbers from Table 5 (31 Oct). VOUT/+VSENSE/COMPLV/CHART/RSET unconnected for current-only use; CHECK the datasheet's unused-pin advice
C326: NOT FITTED.
C327: NOT FITTED.
C328: NOT FITTED.
C329: NOT FITTED.
R302: NOT FITTED. RSET to REFIN, 100× to REFGND
R303: NOT FITTED.
R304: NOT FITTED.
J26: NOT FITTED, 4–20 mA out OUT×, 0 V per channel (active, board powers the loop)

Generated from design.py
Isolated: 6 x 4–20 mA, level and mass-flow ports (AD4112); 6 temperatures (ADS124S08)
EnviroDevelo
Sheet: /Analog inputs and temperatures/
File: io_ana.kicad_sch

Title: CEMS core board: Analog inputs and temperatures

Size: A1	Date: 2026-10-31	Rev: 0.2
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 9/9